YIELD SPREAD MEASURES FOR FIXED-RATE BONDS AND MATRIX PRICING

Learning Outcome

- compare, calculate, and interpret yield and yield spread measures for fixed-rate bonds

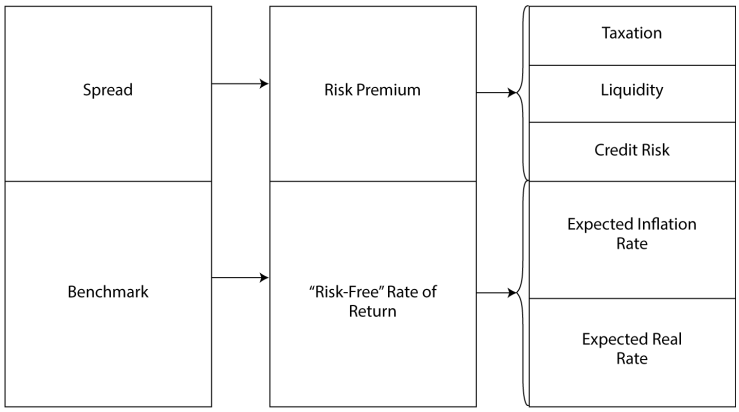
Yield Spreads over Benchmark Rates

In fixed-income security analysis, it is important to understand *why* bond prices and yields-to-maturity change. To do this, it is useful to decompose a yield-to-maturity into a base rate or **benchmark** and an issuer-specific **spread**. For example, an earlier lesson introduced government benchmark bonds as the most liquid securities with the lowest default risk for a specific time-to-maturity in each market. The **yield spread** is the difference between the yield-to-maturity and the benchmark yield.

These two components help investors distinguish between macroeconomic (top-down) and microeconomic (bottom-up) factors that affect the bond price and, therefore, its yield-to-maturity. The benchmark rate captures the top-down factors, while the spread captures the bottom-up factors, as shown in Exhibit 5.

Exhibit 5:

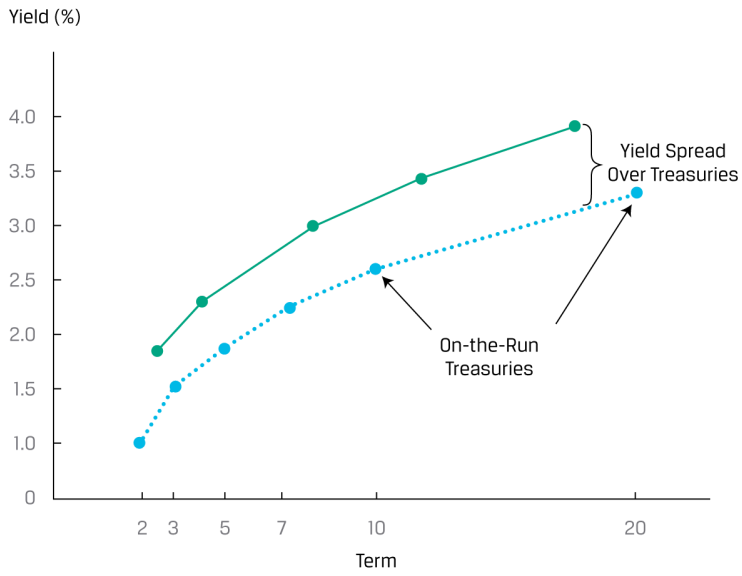
Yield-to-Maturity Components



The most used benchmark rate is the most recently issued government bond, known as an **on-the-run** security. The on-the-run government bond is the most actively traded security and has a coupon rate closest to the current market discount rate for that maturity. That implies that it is priced close to par value. Seasoned government bonds are called **off-the-run**. On-the-run bonds typically trade at slightly lower yields-to-maturity than off-the-run bonds with the same or similar times-to-maturity because of differences in demand for the securities and, sometimes, differences in the cost of financing the government security in the repo market.

The yield spread over a specific benchmark is referred to as the **benchmark spread** and represents the risk premium for the credit and liquidity risks and possibly the tax impact of holding a specific bond. Exhibit 6 shows the relationship between yields-to-maturity for a specific issuer and a benchmark government yield curve.

Exhibit 6:



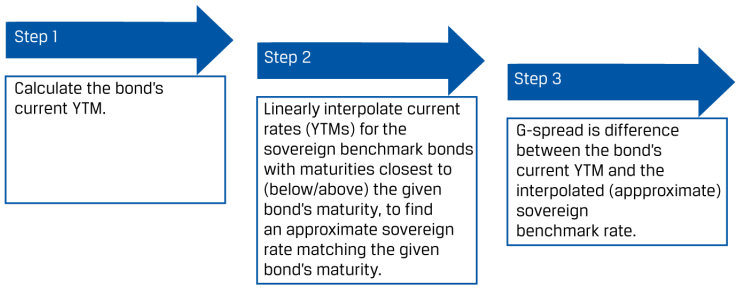
▶ Expand / collapse Extended Description

Spreads allow an analyst to better assess a bond's relative value on a historical and comparative basis by controlling for (excluding) macroeconomic factors, such as a broad increase or decrease in benchmark rates. An analyst might compare the current spread relative to its historical average and its past highs and lows and to other bonds to help identify whether a bond is under- or overpriced on a relative basis.

The yield spread in basis points over an actual or interpolated government bond yield is known as the **G-spread** and represents the return for bearing risks relative to the sovereign bond. If there is no sovereign benchmark bond that matches the maturity of the bond, an approximation process for finding the G-spread must be used, as illustrated in Exhibit 7 and shown in the following example.

Exhibit 7:

Finding G-Spreads When a Comparable Benchmark Bond Does Not Exist



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EXAMPLE 7

Calculating G-Spread without a Comparable Benchmark Bond

Assume that the Russian Federation has issued a 30-year US dollar–denominated sovereign bond based on the following terms:

Issuer:	Russian Federation
Settlement Date:	[T + 5 Business Days]
Maturity Date:	[30 Years from Settlement Date]
Principal Amount:	USD 750 million
Interest:	5.25% fixed semiannual 30/360 basis
Issuance Price:	99.625
Issuance Spread:	200 bp versus current US 30y Treasury
Issuance and Trading:	US Private Placement under Rule 144A
	DTC / Euroclear / Clearstream
Seniority:	Senior Unsecured
Exchanges:	Luxembourg

Six years after the original issuance date, on 15 March 2026, the bond is trading at a price of 123.5 per 100 face value, and an analyst observes that 20-year and 30-year US Treasury yields are currently 2.00% and 2.25%, respectively. Calculate the current G-spread of the Russian Federation bond.

Step 1: Calculate the current yield-to-maturity of the Russian Federation bond. We may solve for r using the general formula for bond price and yield:

$$PV = PMT_1/(1 + r)^1 + PMT_2/(1 + r)^2 + \ldots + (PMT_N + FV_N)/(1 + r)^N.$$

Or we can use the Excel or Google Sheets YIELD function assuming a settlement date of 15 March 2026:

=YIELD(DATE(2026,3,15),DATE(2050,3,15),5.25%,123.5,100,2,0)

= 3.756%.

Step 2: Linearly interpolate the current 20-year (r_{20y}) and 30-year (r_{30y}) US Treasury yields to solve for an approximate 24-year (r_{24y}) US Treasury benchmark:

1. Solve for the weights of the 20-year and the 30-year bonds in the interpolation calculation:

   20-year bond weight = w_{20} = 60% [= (30 – 24)/(30 – 20)].

   30-year bond weight = w_{30} = 40%, or (1 – w_{20}).

   Note that (w_{20} × 20) + (w_{30} × 30) = 24.
2. The 24-year government rate is a weighted average of the 20-year bond rate and the 30-year US Treasury rate using the weights above.

   $r_{24y} = w_{20} \times r_{20y} + w_{30} \times r_{30y}$

   = (60% × 2.00%) + (40% × 2.25%) = 2.10%.

Step 3: The G-spread is the difference between the current Russian Federation bond yield-to-maturity and the 24-year US government rate:

1.656% = 3.756% – 2.10%, so the G-spread is 166 bps.

The yield spread for a bond over the standard swap rate in that currency of the same tenor is known as the **I-spread** or interpolated spread. This type of spread is a comparison against a short-term market-based reference rate and is common practice for pricing and quoting euro-denominated corporate bonds, using a euro interest rate swap as the benchmark. For example, the five-year zero-coupon euro bond issued by the Chinese government at a negative yield-to-maturity from a prior lesson was priced at a rate of “mid-swaps plus 30 bps,” where “mid-swaps” is the average of the bid and offered swap rates.

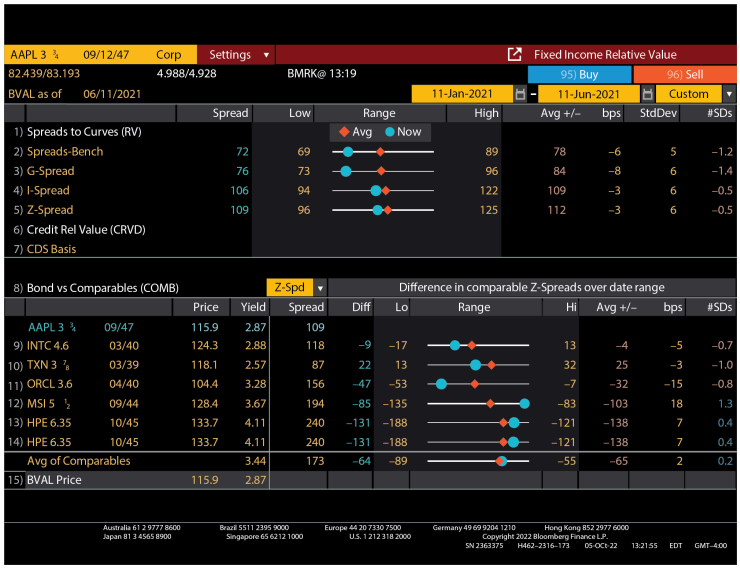
as a measure of a bond's credit risk. Whereas a standard interest rate swap involves an exchange of fixed for floating cash flows based on a floating index, an asset swap converts the periodic fixed coupon of a specific bond to a market reference rate (MRR) plus or minus a spread. If the bond is priced close to par, this conversion approximates the price of a bond's credit risk over the MRR index.

EXAMPLE 8

Yield Spreads Using Bloomberg Fixed-Income Relative Value (FIRV)

The Bloomberg FIRV function for the 3.75% Apple bond that matures on 12 September 2047 shows the following. Identify the yield and yield spread measures.

Figure 1: FIRV Function for 3.75% AAPL Bond Maturing 12 September 2047



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Solution:

The yield spreads are in the top left corner. On the date of analysis:

- The yield spread over a particular Treasury benchmark was 72 bps.
- The G-spread over an interpolated government bond yield was 76 bps. The spread over the benchmark and the G-spread sometimes differ by a few basis points, especially if the benchmark maturity differs from that of the underlying bond.
- The bond's I-spread was 106 bps. An I-spread larger than the G-spread indicates that Treasury yields were slightly higher than swap rates at that time.

The use of these spreads in investment strategies will be covered in more detail later. In general, an analyst will track these spreads relative to their averages and historical highs and lows to identify relative value.

Yield Spreads over the Benchmark Yield Curve

Whereas the G-spread and I-spread are the difference between two single numbers and use the same discount rate (yield-to-maturity) for each cash flow, a different approach is to calculate a constant yield spread over a government (or interest rate swap) spot *curve*, or

benchmark spot rate to make the present value of a bond's cash flows equal its price. In Example 8, the Z-spread for the Apple bond, as quoted using the Bloomberg FIRV function, was 109 bps.

The Z-spread over the benchmark spot curve can be calculated with Equation 4:

$$PV = \frac{PMT}{(1 + z_1 + Z)^1} + \frac{PMT}{(1 + z_2 + Z)^2} + \cdots + \frac{PMT + FV}{(1 + z_N + Z)^N}. \quad 4$$

The benchmark spot or zero rates—z_1, z_2, . . . , z_N—are derived from the government yield curve (or from fixed rates on interest rate swaps), as shown in detail in a later lesson. Z is the Z-spread per period and is the same for all time periods. In Equation 4, N is an integer, so the calculation is on a coupon date when the accrued interest is zero. Sometimes, the Z-spread is called the "static spread" because it is constant (and has zero volatility). In practice, the Z-spread is usually calculated in a spreadsheet using a goal seek function or similar solver function.

The Z-spread is also used to calculate the **option-adjusted spread** (OAS) on a callable bond. The OAS, like the option-adjusted yield, is based on an option-pricing model and an assumption about future interest rate volatility. Then, the value of the embedded call option, which is stated in basis points per year, is subtracted from the yield spread.

OAS = Z-spread – Option value in basis points per year. 5

This important measure is covered in detail later.

Exhibit 8:

Summary of Yield Spreads

Type	Description
G-spread	Government spread, the yield spread in basis points over an actual or interpolated government bond. Used in the US, the UK, Japan, and other jurisdictions.
I-spread	Interpolated spread, the yield spread of a bond over the standard swap rate in the same currency and with the same tenor. Euro-denominated corporate bonds are typically priced vs. a euro interest rate swap benchmark.
Z-spread	Zero-volatility spread, a constant yield spread over a government (or interest rate swap) spot curve used to derive the term structure of credit spreads for an issuer.
Option-adjusted spread	The Z-spread adjusted for the value of an embedded call option.

EXAMPLE 9

The G-Spread and Z-Spread

A 1.5% semiannual coupon, US dollar–denominated bond was issued by PT Indonesia Infrastructure Finance (IIF) three years ago. Currently, the bond has two years remaining to maturity and is trading at a price of 100.45. The two-year, 0.75% semiannual payment government benchmark bond is trading at a price of 100.750.

The government spot rates (stated as effective annual rates) are as follows:

Maturity	Spot Rate
6M	0.127%
12M	0.249%
18M	0.314%

1. Calculate the G-spread for the IIF bond.
2. Calculate the Z-spread for the IIF bond.

Solution:

The G-spread is the yield spread in basis points over an actual or interpolated government bond. The yield-to-maturity for the IIF bond is 1.271%:

$$100.45 = \frac{0.75}{(1+r)^1} + \frac{0.75}{(1+r)^2} + \frac{0.75}{(1+r)^3} + \frac{100.75}{(1+r)^4}.$$
$$r = 0.006357 \text{ YTM} = \text{r} \times 2 = 0.01271.$$

The yield-to-maturity for the relevant government benchmark bond is 0.373%:

$$100.75 = \frac{0.375}{(1+r)^1} + \frac{0.375}{(1+r)^2} + \frac{0.375}{(1+r)^3} + \frac{100.375}{(1+r)^4}.$$
$$r = 0.0018662. \text{ YTM} = \text{r} \times 2 = 0.00373.$$

Therefore, the G-spread is 0.01271 – 0.00373 = 89.8 bps.

Solution:

The Z-spread is a constant yield spread over a government spot or interest rate swap curve.

The Z-spread for the IIF bond is the value of Z that solves the following equation, using the government spot rates provided:

$$PV = \frac{PMT}{(1+z_1+Z)^1} + \frac{PMT}{(1+z_2+Z)^2} + \cdots + \frac{PMT+FV}{(1+z_N+Z)^N}.$$

$$100.45 = \frac{0.75}{(1+0.00127+Z)^1} + \frac{0.75}{(1+0.00249+Z)^2} + \frac{0.75}{(1+0.00314+Z)^3} + \frac{100}{(1+0.003}$$

The Solver add-in for Microsoft Excel finds Z = 0.00268, or 27 bps, by setting the price (sum of present values of cash flows) equal to 100.45 as the objective and Z as the change variable. Please refer to the candidate learning ecosystem online for a spreadsheet demonstrating the calculation.

Question Set

1. Each of the five items below is a risk of investing in fixed-income instruments. For each, identify the component of yield-to-maturity that provides compensation for that risk: benchmark or spread.

   Credit

   Expected Real Rate

   Taxation

   Expected Inflation Rate

   Liquidity

Benchmark	Spread

Hide Solution

Solution:

Benchmark	Spread
Expected Real Rate	Credit Risk

Liquidity Risk

The two components of yield—the benchmark and the spread—allow us to distinguish macroeconomic and microeconomic factors.

The benchmark yield captures the macroeconomic factors: the expected rate of inflation in the currency in which the bond is denominated, general economic growth and the business cycle, foreign exchange rates, and the impact of monetary and fiscal policy. Changes in those factors impact all bonds in the market.

The yield spread captures the microeconomic factors specific to the bond issuer and the bond itself: credit risk, liquidity and trading in comparable securities, and the tax status of the bond.

2. A Mexican corporate non-callable bond matures in three years and has an annual coupon of 5.5%. The government spot rates (stated as effective annual rates) are as follows:

Government Spot Rates	
Maturity	**Spot Rate**
1Y	2.98%
2Y	3.48%
3Y	4.00%

Using the provided spot rates and a Z-spread of 378 bps, calculate the price of the Mexican corporate bond.

Hide Solution

Solution:

The price of the bond is 94.20.

$$PV = \frac{PMT}{(1+z_1+Z)^1} + \frac{PMT}{(1+z_2+Z)^2} + \cdots + \frac{PMT+FV}{(1+z_N+Z)^N}.$$

$$PV = \frac{5.5}{(1+0.0298+0.0378)^1} + \frac{5.5}{(1+0.0348+0.0378)^2} + \frac{105.5}{(1+0.04+0.0378)^3}.$$

$$PV = 94.20.$$

Use the information provided for Questions 3 and 4.

A fixed-income analyst has compiled the following statistics for three corporate bonds issued by companies domiciled in three different countries.

Bond	Yield-to-Maturity	Applicable Swap Rate
A	13.379%	12.419%
B	8.121%	6.527%
C	5.406%	4.184%

3. Which bond most likely has the greatest credit risk?

A. Bond A
B. Bond B
C. Bond C

Hide Solution

Solution:

B is correct. As a “bottom-up” or issuer-specific risk factor, credit risk is a component of a yield spread. To estimate which bond most likely has the

Bond	Yield-to-Maturity	Applicable Swap Rate	I-Spread
A	13.379%	12.419%	96 bps
B	8.121%	6.527%	159 bps
C	5.406%	4.184%	122 bps

4. Assume that the issuers of Bonds A, B, and C are comparable in terms of credit risk and that the bonds have similar liquidity and tax characteristics. Based solely on the information provided, which bond might offer an investor the *least* relative value?

A. Bond A
B. Bond B
C. Bond C

Hide Solution

Solution:

A is correct. If the bonds are judged to be comparable in all material respects, then the least (most) relative value corresponds to the bond with the lowest (highest) spread. At 96 bps, Bond A has the lowest spread and thus the least attractive valuation to an investor.

5. Match the following characteristics to *on-the-run* and *off-the-run* bonds:

- Seasoned government bonds
- Most recently issued government bonds
- Has a coupon rate below or above the current market discount rate for that maturity
- Has a coupon rate closest to the current market discount rate for that maturity
- Trades at slightly lower yields-to-maturity compared to bonds with the same or a similar time-to-maturity
- Trades at slightly higher yields-to-maturity compared to bonds with the same or a similar time-to-maturity

Hide Solution

Solution:

On-the-Run	Off-the-Run
Most recently issued government bonds	Seasoned government bonds
Has a coupon rate closest to the current market discount rate for that maturity	Has a coupon rate below or above the current market discount rate for that maturity
Trades at slightly lower yields-to-maturity compared to bonds with the same or a similar time-to-maturity	Trades at slightly higher yields-to-maturity compared to bonds with the same or a similar time-to-maturity

OPTIONAL UNTESTED SEGMENT

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YAS <GO> integrates multiple fixed income functions, so you can evaluate the pricing, risk, and relative value of a fixed income security in one place. You can review descriptive information, analyze relative value, price a security based on custom inputs, and calculate yields using various market conventions, day types, and compounding methods." -Bloomberg

To explore the YAS function, let's first identify a bond to research. Type SECF into the Bloomberg command line, and press Enter or <GO>. In the top left amber Category dropdown menu, select Fixed Income. Click on one of the bonds listed. Then select 5) YAS from the popup menu. Click on the gray dropdown next to "Yield" and toggle between Maturity (Yield to Maturity) and Worst (Yield to Worst). Note that the features available in YAS vary depending on the security you have loaded for analysis.

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In the practice Qs, how do I solve this? What is a QWE bond? Where do I find the info on it?

The I-spread (in basis points) for the QWE bond is closest to:

A. 46.

B. 80.

C. 87.

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NA

Does anyone know the calculator inputs to get the result from Step 1, Example 7 and Question 2, Example 9 please?

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They solved the Z-spread using excel which we won't have during exam. Will they ask us to calculate Z spread in exam?

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